

GEN ASTRO

Epicycle R2026 Beta

From physics to mission design, optimizations, estimation, and visualization — all in one extensible framework

OPEN MODERN POWERFUL FAST MULTI-PLATFORM

BE UP AND RUNNING IN MINUTES

```
julia> using Pkg
julia> pkg"registry add General https://github.com/GenAstro/GenAstroRegistry.git"
julia> pkg"add Epicycle"

julia> using Epicycle
julia> Epicycle.list_examples()
julia> Epicycle.run_example("Ex_GettingStarted")
```

TRANSPARENCY WHERE IT MATTERS



Open core for design
and analysis

Enterprise capabilities
for operations and
controlled functionality

BEST-IN-CLASS TOOLS



VS Code
Best in class IDE



Cesium
World class 3D
graphics



Plotly
Beautiful data
visualization



Julia
Fast, intuitive
scientific computing

ONE FRAMEWORK MORE THAN PROPAGATION



**Trajectory
Design/Analysis**



Optimal Control
Collocation and
Shooting Methods



**Orbit
Determination**
Batch and sequential



Unified Core
One architecture,
shared physics,
shared derivatives

BUILT FOR WHAT'S NEXT



**Differentiable
by Design**

Enabling
optimization,
estimation, and ML

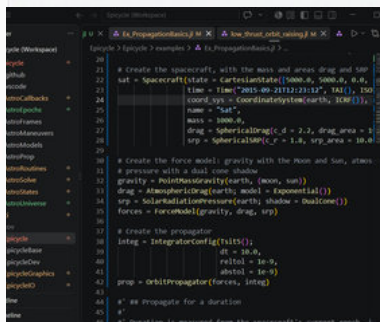


**AI Native
Extensibility**
use LLMs to extend,
analyze, and design



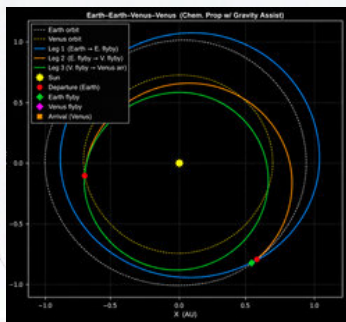
**A Modern
Foundation**

Build from the
ground up on a
modern tech stack



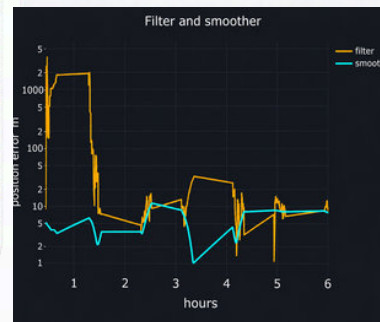
Design

Turn concepts into answers, fast.



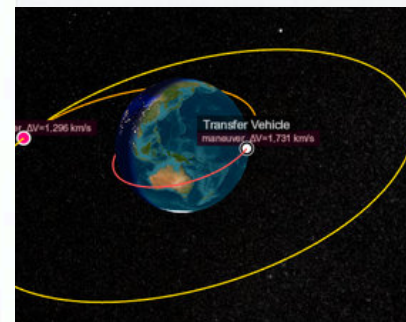
Optimize

Make every design better



Navigate

Design and OD speak the same language



Innovate

Build what your mission needs

SOLVE HARD PROBLEMS FASTER THAN YOU THOUGHT POSSIBLE

A new way to design and navigate space missions—unifying physics, optimization, estimation, and visualization in one open, extensible environment.



A MODERN ASTRODYNAMICS FOUNDATION

ACCURATE. CONSISTENT. DIFFERENTIABLE.

Epicycle provides a unified, high-fidelity foundation for time, states, frames, and the universe. The same physical models are used across propagation, optimization, estimation, and analysis.



TIME

Time scales/formats

TAI, UTC, TT,
TDB, TCG
ISO, Julian Date,
Modified Julian Date

Sub-nanosecond
precision



STATES

12+ representations

Cartesian/Keplerian
Spheric (RA/Dec, AZ/FPFA)

Asymptotes (in/out)

Brouwer mean elements
& more



FRAMES

Precision axes library

Inertial (IAU 2006/2010, FK5)

Planetary body-fixed

Orbit-relative

User-defined frames



UNIVERSE

11 built-in bodies

Sun, planets, Moon & more

SPICE kernel management

EOP file management

User-defined bodies



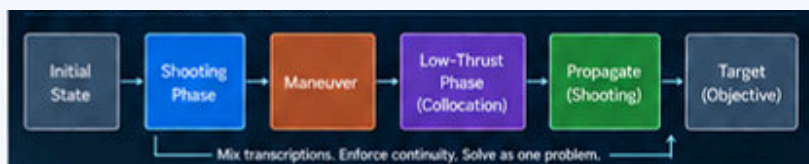
DESIGN & OPTIMIZATION

ONE MISSION • ANY TRANSCRIPTION.

Define the mission objects **once** (e.g. spacecraft, dynamics, controls, variables, constraints, and objectives.)

Solve using parameter optimization, shooting, collocation, or multiple transcriptions **within the same mission**.

Mission Architecture (DAG)



Transcription Methods

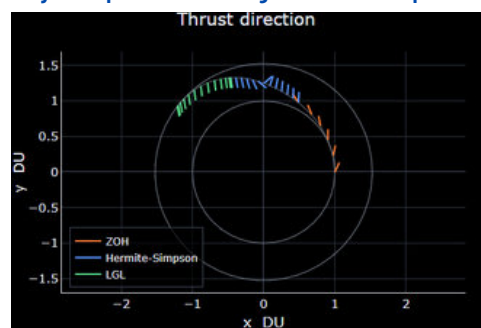


Optimization Capabilities

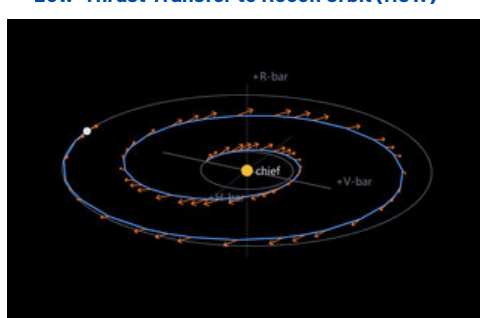
- ✓ Nonlinear programming
- ✓ Single & multiple shooting
- ✓ Collocation (Hermite-Simpson, LGL)
- ✓ Sims-Flanagan
- ✓ Hybrid transcriptions
- ✓ Path & boundary constraints
- ✓ Event-driven problems
- ✓ Analytic and AD partials
- ✓ Scales to large problems

Optimal Control Solved Examples

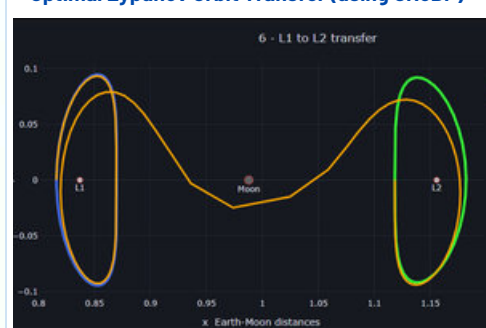
Hybrid Optimization using Three Transcriptions



Low-Thrust Transfer to Recon Orbit (HCW)

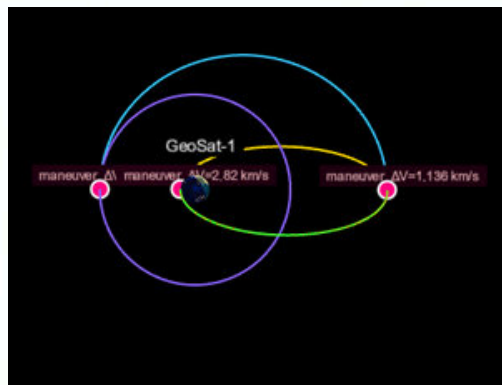


Optimal Lyapunov Orbit Transfer (using CR3BP)

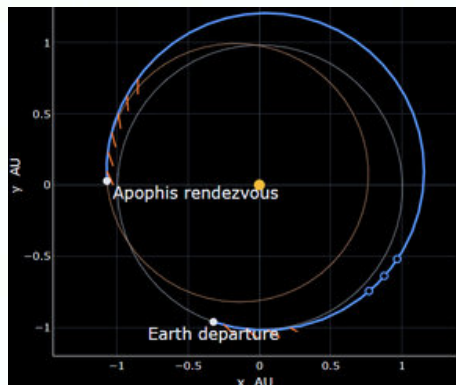


ADDITIONAL EXAMPLE MISSIONS

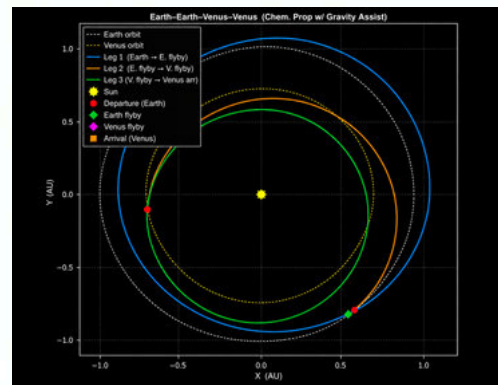
Geosynchronous Earth Orbit Transfer



Apophis Low Thrust Transfer



Earth-Earth-Venus-Venus Optimized Transfer



ORBIT DETERMINATION

ESTIMATION BUILT ON THE SAME PHYSICS ALPHA

Orbit determination is not a separate application. Estimation uses the same states, dynamics, measurements, propagation, and derivative infrastructure as mission design.

SAME PHYSICS.
MORE INSIGHT.
REAL MISSIONS.



BATCH ESTIMATION

Batch least-squares estimation across complete measurement arcs.



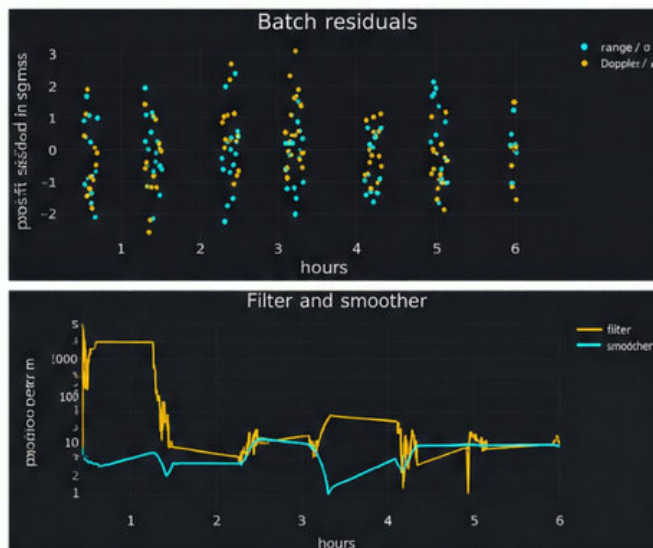
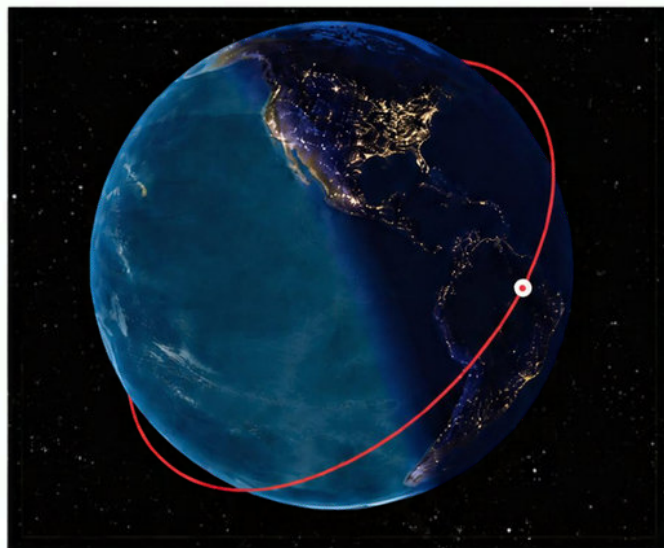
EXTENDED KALMAN FILTER

Sequential state and parameter estimation as measurements arrive.



RTS SMOOTHER

Backward smoothing combines forward-filtered estimates with earlier information.

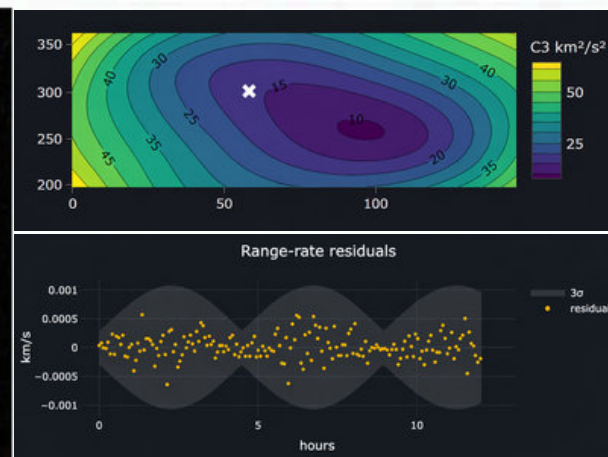
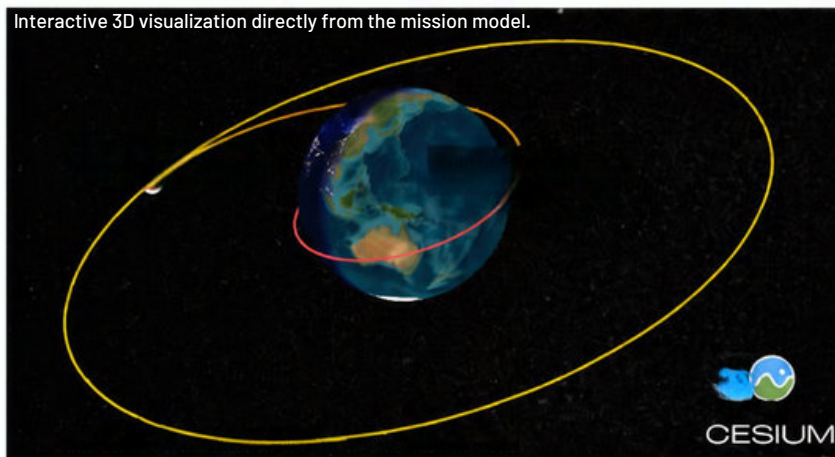


VISUALIZATION

SEE THE MISSION. UNDERSTAND THE DATA.

Visualization is part of the engineering workflow, not something you do at the end.

Interactive 3D visualization directly from the mission model.



Cesium

High-performance interactive 3D mission graphics.



Plotly

Interactive 2D and 3D scientific visualizations.

Explore mission data through dynamic, customizable plots.

Our Model

TRANSPARENCY WHERE IT MATTERS

Most astrodynamics software keeps you guessing or waiting for a release that does what you need. Not Epicycle.

01

Astronomy core is open source.

The foundational astronomy libraries are released under the MT license.

Use, modify and share.

OPEN FOUNDATIONS,
A STRONGER COMMUNITY.



MIT
LICENSE

02

Design and analysis is source available.

Install VS Code, Julia and Epicycle. Start designing solutions today. Want to share your modifications with the community? We partner.

BUILD, EXTEND,
INNOVATE.



GEN ASTRO
SOURCE
AVAILABLE

03

Operational capabilities are enterprise.

High-precision dynamics and precision measurement models in enterprise packages.

Licensing scaled to project scope, not per-seat costs.



ENTERPRISE
LICENSE

Epicycle Tooling

TOOLING BUILT FOR TODAY

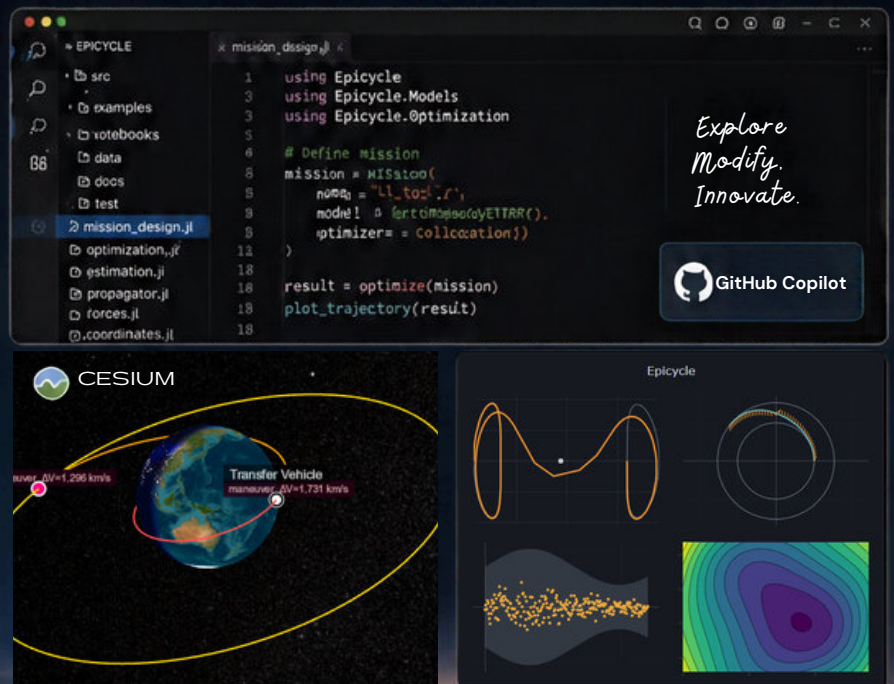
Epicycle runs on best-in-class, open tools and not proprietary components. On Windows, Mac, and Linux.

VS Code provides interactive workspace, native debugging, GitHub Copilot AI Integration, the full extension ecosystem.

Cesium provides STK-quality 3D graphics. Plotty provides beautiful data visualization.

Julia is the language. Fast, powerful scientific computing with Python and MATLAB intuition. Modern architecture from the ground up. ML built in.

You're working in the actual language, not locked in a serialization, scripting, or binding layer.



VS Code
Your IDE
Your workflow.



Cesium
High-performance,
3D visualization.



Plotty
Beautiful,
interactive data.



Julia
Fast-complex scientific
computing. ML built in.

Design, analyze, visualize—all in one environment
with open source tooling you already know, or can quickly learn